

prosail: an R package to simulate canopy reflectance with the coupled model PROSAIL (PROSPECT + SAIL)

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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- [@mmann1123](#)

Submitted: 31 January 2026

Published: unpublished

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Summary

PROSAIL is a vegetation radiative transfer model (RTM) coupling the *leaf optical PRO*PERTIES *SPECTra* (PROSPECT) model, which simulates leaf optical properties (Jacquemoud & Baret, 1990), with the *Scattering by Arbitrarily Inclined Leaves* model (SAIL) which simulates canopy bidirectional reflectance (Berger et al., 2018; Jacquemoud et al., 2009; W. Verhoef, 1984) from the visible to shortwave infrared domains. PROSAIL is a valuable tool for agriculture, ecology, and climate research (Poulter et al., 2023). Inversion procedures allow retrieval of vegetation biophysical properties, including

- leaf area index (LAI)
- leaf chemistry
- the fraction of absorbed photosynthetically active radiation (fAPAR)
- fractional vegetation cover (fCover)

We introduce *prosail*, an R package which provides multiple versions of the model PROSAIL, coupled prospect (Féret & Boissieu, 2024). *prosail* simulates surface reflectance in forward mode for various optical sensors, based on their spectral response function (SRF). It includes functions for model inversion using iterative optimization, and a hybrid approach, combining physical modeling and machine learning (ML) regression.

Statement of need

Vegetation traits mapping and monitoring is crucial to better understand ecosystem functions, carbon, water and energy budgets. Vegetation RTMs describe the interactions between light and vegetation based on their biophysical and chemical properties, through absorption and scattering mechanisms.

The SAIL model is an example of four-stream representations of the radiative transfer equation, including two direct fluxes (incident solar flux and radiance in the viewing direction) and two diffuse fluxes (upward and downward hemispherical flux).

More complex models include the *Soil Canopy Observation of Photosynthesis and Energy fluxes* (SCOPE) (Yang et al., 2021) model, which features layered description of the vegetation, and the *Discrete Anisotropic Radiative Transfer* DART model (Gastellu-Etchegorry et al., 2015), which explicitly describes 3D canopy structure.

Hybrid strategies generating a training dataset for ML algorithms with RTM simulations show multiple advantages (Verrelst et al., 2015): simulations complement or replace in situ

sampling for robust estimation, while ML ensures computation efficiency compared to iterative optimization.

The limited number of input parameters and the computational efficiency of SAIL is interesting for operational applications, despite the main assumption of vegetation as a homogeneous turbid medium: leaves are randomly distributed within the canopy, which is not accurate for row crops and heterogeneous canopies.

State of the field

Various softwares allow hybrid inversion with PROSAIL simulations. The *Sentinel Toolbox Application Platform* (SNAP) includes the Biophysical Processor module (Weiss et al., 2020), combining PROSAIL simulations with an artificial neural network regression model. The sampling design of the training set is described in the *Algorithmic Theoretical Background Document* (ATBD) (Weiss et al., 2020)). Alternative distributions provide more interactive parameterization of the inversion strategy, e.g. the *Automated Radiative Transfer Models Operator* (ARTMO) Matlab toolbox (Caicedo et al., 2014).

prosail includes 4SAIL (W. Verhoef et al., 2007), and 4SAIL2 (Wout Verhoef & Bach, 2007), a two layers version of 4SAIL.

prosail does not intend to provide the same computational efficiency as SNAP. It does not provide a collection of models and methods as comprehensive as those provided with the ARTMO box neither. prosail provides a flexible open source framework to experiment with hybrid inversion procedures, using simple yet fast and efficient training stage, allowing experimenting on training data sampling design, introduction of noise, feature selection over optical sensors. It is suitable for research and development, to identify potential and limitations of inversion strategies. Resulting regression models can be applied on remote sensing data, but it may not be the appropriate software for scaling up operational vegetation monitoring applications.

Alternative PROSAIL implementations are also available in python, Julia and R. Other options are available on [this webpage](#).

Software design

prosail uses prospect for the simulation of leaf optical properties. prosail provides user-friendly and modular functions to simulate vegetation canopy reflectance, and to predict biophysical properties using inversion.

Hybrid inversion is a multi-step procedure requiring simulation of sensor reflectance to train a ML regression algorithm, then applicable to any data source collected from airborne and spaceborne sensors.

- The generation of realistic reflectance simulations requires definition of input parameter sampling strategy, accounting for sensor SRF and uncertainties. prosail provides functions to produce this training dataset as a unique step following standard parameterization, or step by step, in order for users to understand the logic, or adjust each step to their needs.
- A limited number of ML algorithms is currently provided. Additional algorithms will be implemented in future versions.
- Raster processing is handled with the terra (Hijmans, 2025) package, for compatibility with most raster data formats.

- 81 ▪ prosail can be used in combination with [preprocS2](#), a package dedicated to downloading
82 and preprocessing of Earth observation data following the [STAC](#) specification, in order
83 to produce a fully automated image access and processing workflow.

84 Research impact statement

85 prosail has been used in multiple research publications since its early developments ([Hauser
86 et al., 2021](#)), ([Ferreira et al., 2026](#)), ([Féret et al., 2026](#)), ([Kattenborn et al., 2024](#)). It is
87 currently used in multiple research projects, and is actively maintained.

88 Overview

89 PROSAIL simulation in forward mode

90 PROSAIL requires information intrinsic to vegetation :

- 91 ▪ leaf optical properties, either simulated with prospect, or measured experimentally.
92 ▪ LAI
93 ▪ Leaf inclination distribution, defined from different options of *Leaf Inclination Distribution
94 Functions* (LIDF).
95 ▪ foliage hot spot parameter ([Bréon et al., 2002](#); [Kuusk, 1991](#); [Wout Verhoef & Bach,
96 2007](#))

97 PROSAIL also requires information extrinsic to vegetation :

- 98 ▪ sun-observer geometry: sun and observer zenith angles, relative azimuth angle
99 ▪ soil reflectance

100 SAIL produces top-of-canopy bi-hemispherical reflectance factor, directional-hemispherical
101 reflectance factor for solar incident flux, hemispherical-directional reflectance factor in viewing
102 direction and bi-directional reflectance factor.

103 Simulating surface reflectance acquired by a sensor

104 Simulating sensor surface reflectance requires a SRF. A selection of SRF corresponding to
105 multiple sensors is already implemented in prosail, including Sentinel-2, Landsat and MODIS.

106 User-defined sensors require defining central wavelength and full width at half maximum (fwhm)
107 for each band, assuming gaussian response. Exact SRF from external files can also be provided.

108 PROSAIL hybrid inversion

109 Simulating surface reflectance to prepare for regression model training

110 Preparing for training regression models needs defining:

- 111 ▪ a sampling strategy (range, distribution, co-distributions for input parameters)
112 ▪ a level of noise applicable to the simulated surface reflectance
113 ▪ spectral bands used for the training of the regression model

114 The default parameterization of prosail hybrid inversion relies on the production of a surface
115 reflectance LUT compliant with the specifications defined in the ATBD for the Biophysical
116 Processor of the Sentinel toolbox ([Weiss et al., 2020](#)). Each step of the simulation of a training
117 LUT can be defined by user.

118 **Training a ML regression algorithm**

119 The ML strategy implemented in *prosail* is a parsimonious ensemble method based on a
120 bootstrap aggregating (bagging) prediction of biophysical properties from support vector
121 regression (SVR) models. Each SVR model is trained with a limited number of samples,
122 ensuring fast training stage. The predicted value then corresponds to the mean prediction
123 from a set of SVR models. The standard deviation is derived from this ensemble of predictors.
124 However, standard deviation may not provide accurate model uncertainty quantification (Palmer
125 et al., 2022).

126 The default parameterization uses an ensemble of 10 SVR models trained with 200 samples
127 each. The training then requires 2000 samples, to be compared to the 41472 simulations used
128 for training models implemented in SNAP. This parsimonious ensemble method ensures fast
129 training stage and retrieval performances similar to SNAP.

130 **Performing full hybrid inversion**

131 Figure 1 summarizes the workflow applied to perform hybrid inversion. The function
132 `train_prosail_inversion` combines aforementioned steps to simulate surface reflectance
133 and train SVR models. Obtained regression models are then used to estimate vegetation
134 biophysical properties from data tables or raster data.

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Figure 1: Workflow of PROSAIL hybrid inversion implemented in the package prosail.

Application to Sentinel-2 image and comparison with SNAP

Once ML regression models are trained, prosail hybrid inversion can be applied on Sentinel-2 imagery using the function `apply_prosail_inversion`.

Biophysical variables produced with SNAP are compared to those produced with the prosail hybrid inversion. The Sentinel-2 Level-2A product corresponding to tile *30SWJ* acquired on May 13th, 2021 was downloaded from the Copernicus Data Space Ecosystem (CDSE).

LAI, fcover and fAPAR computed from Sentinel-2 images with prosail hybrid inversion are compared to those produced with SNAP in Figure 2.

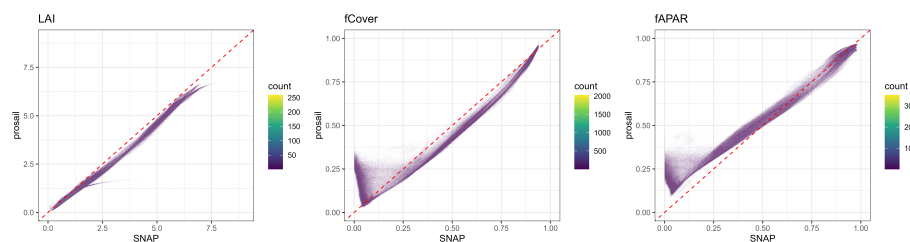


Figure 2: Biophysical properties estimated with *prosail* vs Biophysical properties estimated with SNAP.

The two methods show good consistency, with Pearson correlation coefficient > 0.99 for LAI, and > 0.97 for fCover and fAPAR. Algorithmic differences remain between the two implementations, including: - the version of the PROSPECT model: SNAP uses a version anterior to PROSPECT-D, - the ML algorithm.

Differences in soil properties accounted for in simulations with low LAI may also contribute, despite efforts to reproduce the workflow described in the ATBD. Additional tests performed over croplands and forests showed similar performances.

The availability of open source and fully parameterizable inversion procedures should contribute to improve reproducibility of currently available softwares.

Conclusion

prosail is an R package dedicated to the canopy reflectance model PROSAIL, simulating reflectance from optical sensors based on their spectral response. It includes inversion based on iterative optimization and ML/RTM inversion.

Vegetation biophysical properties estimated with a hybrid inversion are consistent with estimations from SNAP.

prosail hybrid inversion does not intend to be computationally as efficient as SNAP. It is not appropriate for regional to global scale vegetation monitoring. *prosail* hybrid inversion offers a fully adjustable hybrid inversion framework, including an original parsimonious ML regression method, allowing fast and efficient training. It is a valuable tool for experimenting on the potential and limitation of physically-based retrieval of vegetation traits from optical sensors, including satellite missions already operational and in preparation.

Availability

prosail is an open-source software made available under the MIT license. Tutorials are available at <https://jbferet.gitlab.io/prosail/>.

Acknowledgements

This work was supported in part by CNES through the TOSCA project SYLVOSANSAT, by the Agence Nationale de la Recherche (ANR) through the BioCop project (ANR-17-CE32-0001-01), and by government funding managed by ANR under the France 2030 program as part of the PEPR FORESTTT, MONITOR, reference number ANR-24-PEFO-0003. We are grateful to Wout Verhoef for the 4SAIL and 4SAIL2 models. We are grateful to Stéphane Jacquemoud and Frédéric Baret for the initial version of the PROSPECT model.

AI Usage Disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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